



Supplementary Materials for

Coherent evolution of superexchange interaction in seconds-long optical clock spectroscopy

William R. Milner *et al.*

Corresponding authors: William R. Milner, william.milner@colorado.edu; Jun Ye, ye@jila.colorado.edu

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Materials and Methods

The experimental sequence used on our three-dimensional optical lattice clock is detailed in previous publications from this experiment (22, 23, 46). In this section, we briefly summarize our procedure to prepare a nuclear spin-polarized degenerate Fermi gas. This quantum gas is the starting point before lattice loading and clock spectroscopy. First, approximately 100×10^6 atoms are trapped at $\approx$ mK temperatures in our "blue" MOT using the $\Gamma = 2\pi \times 30.5$ MHz, $^1S_0 \leftrightarrow ^1P_1$ transition. Next, our "red" MOT is prepared on the $^1S_0 \leftrightarrow ^3P_1$ transition with linewidth $\Gamma = 2\pi \times 7.5$ kHz. Given the comparatively smaller 3P_1 transition linewidth and thus Doppler temperature, 10×10^6 atoms are laser cooled to 1μ K temperatures in a 1064 nm crossed dipole trap (XODT). With approximately an equal distribution of all 10 nuclear spin states $|^1S_0, m_F = -9/2, m_F = -7/2 \dots m_F = 9/2\rangle$, forced evaporation is employed to cool the atoms to a degeneracy parameter of $T/T_F \approx 0.1$ with 30×10^4 atoms per spin state (23). For clock spectroscopy, a single Zeeman sublevel is desired, requiring nuclear-spin polarization. For polarization, we apply a spin dependent potential detuned from the 3P_1 transition where the polarizability is negative for all sublevels except $|m_F = 9/2\rangle$. Including an initial optical pumping stage, we prepare a nuclear spin polarized gas in $|^1S_0, m_F = -9/2\rangle$ with temperature $T/T_F \approx 0.2$ (46). We note that the lattice lifetime depicted in Fig. S1 is 108(5) s, much longer than the Ramsey dark times probed in this work.

To confine the atoms in our 3D lattice, we adiabatically ramp all lattice beams in three 150 ms steps. Starting at $0 E_R$, we ramp to $2.5 E_R$, then $10 E_R$, before ramping to our final trap depth V_F for clock spectroscopy. For measurements with $V_F \leq 10 E_R$, we ramp $0E_R \rightarrow 2.5E_R \rightarrow V_F$ then hold. To prepare for clock spectroscopy in our magic wavelength lattice, we ramp off the XODT trap over 100 ms, while leaving the lattice depths at V_F .

The superexchange coupling strength $J_{SE}(j)$ depends on the site-to-site energy shift ΔE_j between atoms on lattice sites j along the clock axis z . Calculating ΔE_j requires determining the atom's location in the lattice Gaussian confinement during clock spectroscopy. We go step-by-step through the loading procedure, considering the trapping potential at each step, to arrive at the final confinement.

First, we consider the trapping potential in the XODT before lattice loading. The XODT potential including the gravitational tilt is $U_{XODT}(j)/\hbar = (1/2 m\omega_{XODT}^2 a^2)(j - j_{XODT})^2 + mga_j$, where j is a dimensionless position in units of the lattice spacing, j_{XODT} is the center position of the XODT beam, and $\omega_{XODT} = 2\pi \times 250$ Hz, thus $U_{XODT}(j)/\hbar = (44.7 \text{ Hz})(j - j_{XODT})^2 + (873.1 \text{ Hz})j$. The bottom of this displaced parabola is at j_s , where the atomic density distribution is centered.

Next, we consider the potential of the optical lattice. We align the center of the lat-

tice beams to the atom cloud in the XODT. Thus, the center of the transverse lattice beams is located at j_s . After ramping down the XODT beams, the offset of the vertical lattice potential, generated by the two transverse lattice beams, is therefore $U_{\text{lat}} = -2V_{\perp} \exp(-2a^2(j - j_s)^2/w^2) + mga j$, where w is the transverse lattice beam waist radius extracted in Fig. S3. Thus, the minimum of U_{lat} is at $j < j_s$. Because we turn off the 1064 nm trap in a deep lattice where tunneling is frozen, the atom position is centered at j_s throughout the Ramsey spectroscopy. Therefore, the cloud is significantly displaced from the center of the lattice potential, as depicted in Fig. 4C. At shallow transverse lattice depths where the Gaussian confinement is weak, Wannier-Stark peaks at approximately mga are observed as seen in Fig. S2. As the transverse confinement, and thus the inhomogeneity, is increased these peaks are broadened.

The XODT trap frequency ν_{XODT} is calibrated via sloshing measurements, where the atoms are displaced in the dipole trap and their spatial oscillations at ν_{XODT} are measured. We do a similar measurement to determine the Gaussian confinement from the lattice beams. Here, the atoms are loaded from the XODT into a 2D lattice formed by both transverse lattice beams. As depicted in Fig. S3, the oscillation frequency is plotted as a function of transverse trap depth $V_{\perp}$.

For the measurements in Fig. 4D, we displace the position of the atoms in the lattice potential to locally change ΔE_j . This requires precise adjustment of the density distribution at the μm level. To accomplish this, we use our horizontal imaging system with imaging parameters in (46). We used a commercial, piezo actuated mirror that moves the XODT beam to adjust the center position of the atomic density distribution.

Our saturated imaging procedure using our high-resolution ($\text{NA} = 0.2$) vertical imaging system is detailed in (46). We extract the contrast of our atomic ensemble using imaging spectroscopy where the local excitation fractions in two spatially separated regions are compared. Ellipse fitting is used, where parametric plots in regions P_1, P_2 trace an ellipse $= \frac{1}{2} + \frac{C}{2} \cos(2\pi f_{1,2}T + \phi_0)$ (48). The differential frequency shift $f_{1,2}$ between spatial regions of interest following our XY8 decoupling sequence is consistent with zero. This is illustrated in Fig. 2B, where the parametric plots show a straight line with no opening angle. In regions-of-interest much larger than P_2 , we observe that the fraction of atoms participating in superexchange is reduced due to the reduced filling fraction in the thermal wings of the cloud as shown in Fig. S4. We use jackknifing to determine the 1σ (standard deviation) contrast error bars. For n excitation fraction measurements at a given dark time T , we first determine the contrast $\bar{C}$ using all data points. Next, we cycle through all n datapoints and recompute the contrast $C_{\neq i}$ excluding the i -th data point. Summing all contributions we estimate the uncertainty: $\text{Var}(C) = \frac{n-1}{n} \sum_i^n (\bar{C} - C_{\neq i})^2$

Fermi-Hubbard Hamiltonian and superexchange model

In our experiment operating under the conditions $V_z \ll V_\perp$, we ignore tunneling in the transverse directions and consider dynamics only along the z direction, which are generated by superexchange processes between adjacent sites. To model the dynamics, we start from the two-site Fermi-Hubbard Hamiltonian with a energy tilt ΔE between sites, with a single atom populating each site,

$$\hat{H}_{\text{FH}} = -t_z \sum_{\sigma \in \{g,e\}} (\hat{c}_{0,\sigma}^\dagger \hat{c}_{1,\sigma} + H.c.) + U \sum_{j \in \{0,1\}} \hat{n}_{j,e} \hat{n}_{j,g} + \frac{\Delta E}{2} (\hat{n}_1 - \hat{n}_0). \quad (\text{S1})$$

Here, $\hat{c}_{j,\sigma}^\dagger (\hat{c}_{j,\sigma})$ creates (annihilates) a fermion on site j with spin σ in the lab frame. We define $\hat{n}_{j,\sigma} = \hat{c}_{j,\sigma}^\dagger \hat{c}_{j,\sigma}$, and $\hat{n}_j = \hat{n}_{j,e} + \hat{n}_{j,g}$. In the Mott-insulating limit $U \gg t_z$, the double-occupied states are separated by a large energy gap $\sim U$, which allows for restriction of dynamics in the single-occupied states via second-order perturbation theory. The resulting effective Hamiltonian describing the superexchange interaction is

$$\hat{H}_{\text{eff}} = J_{\text{SE}} \left(\hat{\mathbf{s}}_0 \cdot \hat{\mathbf{s}}_1 - \frac{1}{4} \right), \quad (\text{S2})$$

where

$$J_{\text{SE}} = \frac{4t_z^2 U}{U^2 - \Delta E^2}. \quad (\text{S3})$$

Here the spin operators are defined as $\hat{\mathbf{s}}_j = (\hat{s}_j^x, \hat{s}_j^y, \hat{s}_j^z) = \sum_{\alpha\beta=\{e,g\}} \hat{c}_{j,\alpha}^\dagger \boldsymbol{\sigma}_{\alpha\beta} \hat{c}_{j,\beta}/2$, where $\boldsymbol{\sigma} = (\sigma^x, \sigma^y, \sigma^z)$ is the vector of Pauli matrices. Expanding $\hat{\mathbf{s}}_0 \cdot \hat{\mathbf{s}}_1 = (\hat{\mathbf{s}}_0 + \hat{\mathbf{s}}_1)^2/2 - 3/4$, we obtain triplet states ($|d\rangle \equiv |g, g\rangle$, $|u\rangle \equiv |e, e\rangle$, $|t\rangle \equiv (|g, e\rangle + |e, g\rangle)/\sqrt{2}$) with zero energy, and the singlet state ($|s\rangle \equiv (|g, e\rangle - |e, g\rangle)/\sqrt{2}$) with energy $-J_{\text{SE}}$ with respect to the triplet states.

The oscillation of the Ramsey contrast can be understood based on the time evolution of the initial state generated by the first Ramsey pulse. This state has a spiral phase $e^{ij\varphi}$ imprinted by the clock laser with Rabi frequency Ω , $\hat{H}_{\text{clock}}(\theta)/\hbar = \frac{1}{2} \sum_j (|\Omega| e^{i\theta} \times \hat{s}_j^\dagger e^{ij\varphi} + H.c.)$, where θ controls the rotation axis. We consider the initial state generated by the first Ramsey pulse with $|\Omega|T_1 = \pi/2$ and $\theta = \pi/2$,

$$\begin{aligned} |\psi_{\text{init}}\rangle &= e^{-i\hat{H}_{\text{clock}}(\theta=\pi/2)T_1/\hbar} |g\rangle_0 \otimes |g\rangle_1 \\ &= \frac{1}{\sqrt{2}} (|g\rangle_0 + |e\rangle_0) \otimes \frac{1}{\sqrt{2}} (e^{-i\varphi/2} |g\rangle_1 + e^{i\varphi/2} |e\rangle_1) \\ &= \frac{1}{2} \left[e^{-i\varphi/2} |d\rangle + e^{i\varphi/2} |u\rangle + \sqrt{2} \cos(\varphi/2) |t\rangle + i\sqrt{2} \sin(\varphi/2) |s\rangle \right]. \end{aligned} \quad (\text{S4})$$

Following the pulse, the dynamics in the dark time can be described by the system evolving under the superexchange Hamiltonian for time T , which amounts to the singlet state $|s\rangle$ acquiring a phase $e^{iJ_{\text{SE}}T/\hbar}$.

$$\begin{aligned} |\psi_f(T)\rangle &= e^{-i\hat{H}_{\text{eff}}T/\hbar} |\psi_{\text{init}}\rangle \\ &= \frac{1}{2} \left[e^{-i\varphi/2} |d\rangle + e^{i\varphi/2} |u\rangle + \sqrt{2} \cos(\varphi/2) |t\rangle + ie^{-iJ_{\text{SE}}T/\hbar} \sqrt{2} \sin(\varphi/2) |s\rangle \right]. \end{aligned} \quad (\text{S5})$$

For simplicity, we ignore all the echo pulses during the dark time. Then we apply the second Ramsey pulse with $|\Omega|T_2 = \pi/2$, $\theta = \pi/2$ and get the final state,

$$|\psi_f\rangle = e^{-i\hat{H}_{\text{clock}}(\theta=\pi/2)T_2/\hbar} |\psi_f(T)\rangle, \quad (\text{S6})$$

for which the Ramsey contrast is given by $C = 2|\langle\psi_f|\hat{s}^z|\psi_f\rangle|/N$ (here $N = 2$) with $\hat{s}^z = \hat{s}_1^z + \hat{s}_2^z$, which can be written analytically as

$$C(T) = \left| \cos^2\left(\frac{\varphi}{2}\right) + \sin^2\left(\frac{\varphi}{2}\right) \cos(J_{\text{SE}}T/\hbar) \right|. \quad (\text{S7})$$

From this form we see that the Ramsey contrast will undergo oscillatory dynamics with frequency set by the superexchange matrix element J_{SE} .

Beyond the two-site case, for a chain of $j = 1 \dots L > 2$ adjacent occupied sites along z , we must generalize Eq. (S2) to a many-body Hamiltonian. For this it is convenient to work in a “spiral” frame, where the initial state following the first Ramsey pulse is uniform (all atoms in the same superposition state) and the site-dependent laser phase φ is absorbed into the spin operators across the lattice, $\hat{s}_j^\pm = \hat{s}_j^\pm e^{\pm i j \varphi}$, $\hat{s}_j^Z = \hat{s}_j^Z$. This transformation leads to the following 1D spin-spin interaction Hamiltonian (41),

$$\begin{aligned} \hat{H}_{\text{SE}} &= \sum_{j=1}^{L-1} J_{\text{SE}}(j) \left[\cos(\varphi) (\hat{s}_j^X \hat{s}_{j+1}^X + \hat{s}_j^Y \hat{s}_{j+1}^Y) + \hat{s}_j^Z \hat{s}_{j+1}^Z + \sin(\varphi) (\hat{s}_j^X \hat{s}_{j+1}^Y - \hat{s}_j^Y \hat{s}_{j+1}^X) \right], \\ J_{\text{SE}}(j) &= \frac{4t_z^2 U}{U^2 - \Delta E_j^2}, \end{aligned} \quad (\text{S8})$$

where ΔE_j and $J_{\text{SE}}(j)$ are the local potential difference and superexchange interaction strength between sites $j, j+1$ respectively. These are not uniform, as the local potential differences depend on both gravity and the Gaussian profile of the lattice beams, the latter of which varies over the cloud. For lattice sites $j, j+1$ along z we have $\Delta E_j = \frac{1}{2} m \omega_{\text{lat}}^2 a^2 [(j+1-j_0)^2 - (j-j_0)^2]$, where j_0 is the bottom of the lattice confining potential.

The first three terms in this many-body Hamiltonian act as an XXZ Hamiltonian with spin anisotropy $\sim \sec(\varphi)$, which induces contrast decay. The last two terms are a Dzyaloshinskii-Moriya (DM) type interaction, which breaks exchange symmetry due to the chirality of the imprinted clock laser phase; since the interaction strengths $J_{\text{SE}}(j)$ are inhomogeneous, this DM term will also generate contrast decay, as each atom feels an unequal force from its left and right neighbours.

The initial state in this "spiral" frame reads,

$$|\psi_{\text{init}}\rangle = e^{-i\frac{\pi}{2}\sum_j \hat{s}_j^Y} \bigotimes_j |\downarrow\rangle_j = \bigotimes_j \frac{1}{\sqrt{2}}(|\downarrow\rangle_j + |\uparrow\rangle_j), \quad (\text{S9})$$

where $\downarrow, \uparrow$ correspond to the g, e states in the "spiral" frame. The chain then undergoes time-evolution under the Hamiltonian as before,

$$|\psi_f(T)\rangle = e^{-i\hat{H}_{\text{SE}}T/\hbar} |\psi_{\text{init}}\rangle, \quad (\text{S10})$$

potentially interspersed with the dynamical decoupling pulses if they are included in the theoretical modelling. For instance, a single echo pulse would read $|\psi_f(T)\rangle = e^{-i\hat{H}_{\text{SE}}T/(2\hbar)} e^{-i\pi\sum_j \hat{s}_j^X} e^{-i\hat{H}_{\text{SE}}T/(2\hbar)} |\psi_{\text{init}}\rangle$, while an XY8 sequence instead applies eight pulses about different axes as depicted in the main text Fig. 2A. After time-evolving the state, the final Ramsey pulse is performed,

$$|\psi_f\rangle = e^{-i\frac{\pi}{2}\sum_j \hat{s}_j^Y} |\psi_f(T)\rangle. \quad (\text{S11})$$

The contrast of the chain is obtained by measuring the excited state fraction as before,

$$N_{\text{exc}}(T) = \langle \psi_f | \left[\sum_j \left(\hat{s}_j^Z + \frac{1}{2} \right) \right] | \psi_f \rangle, \quad (\text{S12})$$

$$C(T) = \frac{1}{L} N_{\text{exc}}(T),$$

where the $1/L$ prefactor ensures normalization $C(T = 0) = 1$. If there are multiple independent chains, their contributions to $N_{\text{exc}}(T)$ are first summed together, then the contrast $C(T)$ is normalized by the total number of particles at the end.

A single site $L = 1$ has unity contrast $C = 1$. Chains with few sites will exhibit persistent oscillations of contrast, whereas chains with many sites will undergo decay, with revivals only occurring on timescales $\sim 1/L$. Inhomogeneity in the superexchange couplings $J_{\text{SE}}(j)$ will also wash out revivals or oscillatory dynamics at longer times. The contrast dynamics from many summed, disordered chains thus generally exhibits only one

or two oscillations before saturating to a constant value determined by how many of the chains were isolated single sites with $L = 1$.

All of the theoretical simulations presented in the main text Fig. 3 and Fig. 4 model the contrast decay according to the above theory, employing various levels of complexity for the parameters and solution of the equations. In order of appearance, we have:

- Figs. 3A, 3B numerically solve the full many-body Hamiltonian evolution and average over experimental conditions, using independently calibrated parameters to determine the superexchange couplings $J_{\text{SE}}(j)$ and chain lengths L , and including the dynamic decoupling pulses. This approach is faithful to the experimental conditions, but numerically expensive. The details are in Supplementary *Full cloud simulations* below.
- Figs. 3C, 4E also average the superexchange-induced contrast decay over experimental conditions, but only consider contributions from two-site pairs, and neglect the decoupling pulses. This averaging is more efficient as one can average the analytically-computed contrast in Eq. (S7) rather than numerically time-evolving the Hamiltonian. The details are in Supplementary *Hereustic averaging of superexchange dynamics* below.
- Fig. 3D uses a more general treatment agnostic to experimental conditions. This approach again solves the full many-body Hamiltonian evolution, but averages the couplings and chain lengths from well-motivated generic probability distributions rather than experiment-specific values. In particular the couplings $J_{\text{SE}}(j)$ are drawn from a Gaussian distribution, while the lengths L are drawn from a Poisson distribution. The details are in the Supplementary *General averaging* below.

Full cloud simulations

To model the experimental contrast decay in Fig. 3A and Fig. 3B, we average over the 3D distribution using calibrated experimental parameters. We first consider the initialization of the atomic cloud in the 3D lattice + XODT trap. Assuming that the XODT trap depth is equal along the x, y directions, the system has cylindrical symmetry in the $x - y$ plane, and all vertical z -direction tubes at fixed horizontal radius $r_{xy} = \sqrt{x^2 + y^2}$ from the trap center have the same physics.

Within a given lattice tube at fixed r_{xy} , we assume that the loading procedure yields an

ensemble of atoms in a discrete Fermi-Dirac distribution,

$$\begin{aligned} P(r_{xy}, z) &= \frac{1}{e^{(E(r_{xy}, z) - \mu)/(k_B T_{lat})} + 1}, \\ E(r_{xy}, z) &= \frac{1}{2} m \omega_{XODT, \perp}^2 a^2 r_{xy}^2 + \frac{1}{2} m \omega_{XODT}^2 a^2 z^2, \end{aligned} \quad (\text{S13})$$

where z is the integer lattice site index along the vertical tube direction, T_{lat} is the temperature in the lowest band of the 3D lattice, k_B is Boltzmann's constant, μ is a chemical potential offset, and $E(r_{xy}, z)$ is the local potential energy from the XODT, with $(\omega_{XODT, \perp}, \omega_{XODT})/(2\pi) = (136, 250)$ Hz the horizontal and vertical trapping frequencies of the XODT (the actual XODT trap in the experiment has slightly unequal trapping frequencies along the x and y horizontal directions, so we take $\omega_{XODT, \perp}$ as an average of the two for simplicity). This approach employs the local density approximation, neglecting any tunneling effects or band excitations. We omit gravity by assuming that it is incorporated into the definition of the z -direction potential; i.e. the minimum of the XODT potential and gravity is at $z = 0$, and the overall center of the potential is at $r_{xy} = z = 0$ during the loading process.

The total number of atoms is given by the integral of the probability density,

$$N = \int_0^\infty dr_{xy} r_{xy} \int_{-\infty}^\infty dz P(r_{xy}, z) = - \left(\frac{2\pi k_B T_{lat}}{m a^2} \right)^{3/2} \frac{1}{\omega_{XODT, \perp}^2 \omega_{XODT}} \text{Li}_{3/2} \left[-e^{\mu/(k_B T_{lat})} \right], \quad (\text{S14})$$

where $\text{Li}_n[\cdot]$ is the polylogarithm function. For fixed trapping frequencies and atom number N , the gas is described by just the temperature T_{lat} , while the chemical potential offset μ is fixed via the above equation. Atom numbers vary between different experiments, so we take these changes into account for the calculations compared to different data sets.

For each horizontal trap radius, r_{xy} , a set of initial particle distributions is randomly sampled via the Fermi-Dirac distribution. Each such distribution is a set of occupied sites $\mathbb{P} = \{z\}$. From this sampling we identify all continuous, uninterrupted chains of atoms. For example, if we sampled particles at sites $\mathbb{P} = \{-3, -2, -1, 4, 5, 9\}$, we identify uninterrupted chains $(-3, -2, -1)$, $(4, 5)$, (9) , assuming that atoms separated by one or more empty sites do not talk to each other.

For each uninterrupted chain, the quantum dynamics are solved by time-evolving the Hamiltonian including the XY8 decoupling pulses, as described in the *Fermi-Hubbard Hamiltonian and superexchange model section of the Supplementary Materials*. The sampling from the Fermi-Dirac distribution is centered about the XODT potential minimum j_s , while the local energy differences ΔE_j that enter into the superexchange couplings are determined by the lattice potential minimum j_0 via $\Delta E_j = \frac{1}{2} \omega_{lat}^2 a^2 [(j+1-j_0)^2 - (j-j_0)^2]$. Note that to avoid numerical issues in places where the superexchange denominator

$U - \Delta E_j$ vanishes, we also bound the magnitude of $J_{\text{SE}}(j)$ by the tunneling rate t_z . While there can be beyond-superexchange physics in such places, we assume such effects are localized to a few sites and do not affect the overall signal.

The contributions of each uninterrupted chain in the set $\mathbb{P}$ to the final excited state fraction are solved separately, then summed together,

$$N_{\text{exc},r_{xy}}(T) = \sum_{\mathbb{P}} N_{\text{exc}}(T). \quad (\text{S15})$$

Many thermal trajectories (~ 100) with different initial particle distributions $\mathbb{P}$ are sampled, and an average is taken for each time step. This calculation is repeated for discrete horizontal trap radii $r_{xy} = 0 \dots 16$ sites corresponding to a horizontal cylinder of roughly $6\mu\text{m}$ radius, which is the experimentally imaged region P_1 . The contributions of tubes at different r_{xy} are then averaged, weighted by the circumference $2\pi r_{xy}$, which is the number of tubes at that radius,

$$N_{\text{exc,total}}(T) = \sum_{r_{xy}} (2\pi r_{xy}) N_{\text{exc},r_{xy}}(T). \quad (\text{S16})$$

The total contrast is then obtained by normalizing,

$$C(T) = \frac{N_{\text{exc,total}}(T)}{N_{\text{exc,total}}(0)}. \quad (\text{S17})$$

Finally, since the experiment also finds a slower decay on timescale $\sim T_2$ measured in Fig. 2C that stems from physics beyond superexchange, we normalize the resulting contrast obtained from the numerical simulation of the superexchange Hamiltonian by a further factor $C(T) \rightarrow C(T)e^{-T/T_2}$.

This calculation is repeated for a range of different initial temperatures T_{lat} . Fig. S5 shows the resulting contrast for different transverse lattice depths $V_{\perp}$ probed in the experiments shown in Fig. 3. The corresponding calibrated parameters used in these simulations are given in Table S1. We find agreement for all but the shallowest $V_{\perp}$ (for which in-plane interactions start to play a role). Higher temperatures lead to a larger long-time average value for the contrast, but do not strongly affect the oscillatory profile. This is because the temperature only determines how likely one is to find atoms next to each other in longer chains, but does not directly change the ensuing quantum dynamics of the chains. The main text shows simulations with the best-fitting temperature of 350(14) nK for Fig. 3A and 322(17) nK for Fig. 3B, which is in good agreement with the measurements. The shaded region on the theory curves corresponds to a combined uncertainty of the fitted temperature and ± 2 s for the T_2 in the exponential normalization factor (in line with the T_2 measurement uncertainty).

Heuristic averaging of superexchange dynamics

While averaging over all chain lengths is numerically expensive, at the regime of lattice filling in our experiment a large portion of the experimental signal comes from superexchange between pairs of sites. So we can heuristically generalize Eq. (S7) to average over all possible local one-atom and two-atom chains in our system rather than numerically solving the full many-body Hamiltonian. We define $n(j)$ as the local filling fraction of lattice sites labelled by j , such that the total atom number in each vertical tube is $N_{\text{tube}} = \sum_j n(j)$. In the following, two adjacent lattice sites along z direction are considered but all quantities are given with respect to individual lattice sites. The probability to have only one atom in these two lattice sites is $p^{(1)}(j) = n(j)[1 - n(j)]$, and the probability to have two atoms is $p^{(2)}(j) = [n(j)]^2$. Here, the magnetization of a one-atom chain is $\langle \tilde{s}^{z,(1)}(j) \rangle = \langle \psi_f | \hat{s}^z | \psi_f \rangle = 1/2$, and in a local two-atom chain $\langle \tilde{s}^{z,(2)}(j) \rangle = \{\cos^2(\varphi/2) + \sin^2(\varphi/2) \cos[J_{\text{SE}}(j)T/\hbar]\}/2$ (see Eq. (S7)) and thus the average magnetization per size is $\langle \tilde{s}^z(j) \rangle = \sum_k p^{(k)} \langle \tilde{s}^{z,(k)} \rangle$. We then average over all spatial positions to obtain the Ramsey contrast

$$\begin{aligned} C(T) &= \frac{1}{N_{\text{tube}}} \left| \sum_j 2 \langle \tilde{s}^z(j) \rangle \right| \\ &= \left| 1 - \sum_j \frac{[n(j)]^2}{N_{\text{tube}}} \sin^2\left(\frac{\varphi}{2}\right) + \sum_j \frac{[n(j)]^2}{N_{\text{tube}}} \sin^2\left(\frac{\varphi}{2}\right) \cos(J_{\text{SE}}(j)T/\hbar) \right|. \end{aligned} \quad (\text{S18})$$

By fitting a function of the form $Ae^{-T/\tau_{\text{osc}}} \cos(\bar{J}_{\text{SE}}T/\hbar)$ to the oscillatory term in $C(T)$ we obtain the oscillation frequency $\bar{J}_{\text{SE}}/\hbar$, which is the basis for the solid red line in Fig. 4E. The error bands are derived from the uncertainty of this fit parameter. The averaging of the filling fraction along the vertical direction is done in the same manner as the Supplementary *Full cloud simulations*. We choose a temperature of ~ 350 nK to roughly match the peak-to-peak oscillation amplitude of $2 \sin^2(\varphi/2) \sum_j [n(j)]^2 / N_{\text{tube}} \sim 0.7$ observed in most measurements for Figs. 4D or e.g. 3B. The average atom numbers for the data in Figs. 3 and 4 are different, so we choose $N = 20000$ and $N = 30000$ for the calculations in Fig. 3 and Fig. 4, respectively.

Note that the above averaging must sample over the non-uniform superexchange rate $J_{\text{SE}}(j)$, for which the formula in Eq. (S3) can diverge when the denominator vanishes. Furthermore, the on-site interactions have a weak dependence on the vertical direction due to the beam waist of the transverse x, y confinement, hence $U \rightarrow U_j$. To avoid both of these issues, we calculate $J_{\text{SE}}(j)$ with an independent diagonalization of Eq. (S1) at each pair of sites $j, j+1$.

Going beyond this treatment, additional interaction and tunneling terms can be present in the Fermi-Hubbard Hamiltonian due to wavefunction overlap with adjacent sites. We

identify the main contributions to be bond-charge type effects (57) and an admixture of higher bands due to the gravitational tilt. The former bond-charge interactions are those with interactions between adjacent sites and additionally an exchange of the particles. This can be cast into the form of a tunneling term, thus directly correcting the tunneling energy $t'_z = t_z + \Delta t_z$ with

$$\Delta t_z = -\frac{4\pi\hbar^2 a_{eg}^-}{m} \int d^3x \psi_0^3 \psi_1, \quad (\text{S19})$$

where $\psi_j = \psi(x, y, z - ja)$ describes the ground band Wannier function ψ at lattice site j . For lattice depths of $V_z = 17.4 E_R$ and $V_\perp = 44.9 E_R$ we obtain $\Delta t \approx h \times 1.2 \text{ Hz}$, which corresponds to an increase of about 8% with respect to the bare value of $t_z \approx h \times 14.2 \text{ Hz}$. The bond-charge corrections Δt_z are taken into account for the blue line in Fig. 3C.

The latter gravity-induced corrections to the Wannier functions can also be computed, yielding additional corrections to the tunneling energy on the order of $\sim 10\%$. However, the exact calculation of the full contribution remains challenging because we estimate that all higher bands would be needed to be taken into account for a faithful quantification (21). Because these effects are barely above our experimental uncertainty we are mostly neglecting these corrections in this work.

General averaging

For main text Fig. 3D, rather than fitting different temperatures to different lattice depth data, we provide a more general theoretical prediction that does not rely on explicit experimental conditions. We randomly sample a large number of chains with lengths L drawn from a Poisson distribution $P(\lambda)$ with Poisson parameter $\lambda < 1$, appropriate for an initial thermal distribution. The coupling strengths $J_{\text{SE}}(j)$ in each chain are drawn from a Gaussian distribution of mean $\bar{J}_{\text{SE}}$ and standard deviation $\epsilon \bar{J}_{\text{SE}}$, with ϵ meant to capture inhomogeneity in the superexchange interactions due to the local potential differences.

As ϵ increases, the contrast oscillations reduce in amplitude to the profile observed in the experiment. The curve in Fig. 4D of the main text shows the prediction for Poisson parameter $\lambda = 0.25$ and $\epsilon = 0.4$. This curve is also scaled by a factor of $e^{-\frac{1}{5}T\bar{J}_{\text{SE}}/h}$ to account for an additional decay of atomic coherence on a longer timescale as observed in the measurements, using an effective lifetime of five superexchange cycles.

Investigation of atomic coherence close to the 1D limit

The coherence lifetime measurements in Fig. 2C suggest interesting physics that manifests in the near-1D regime ($V_\perp \approx 0$). In this regime there are strong interactions within each horizontal x - y plane of the lattice; however, since there is no in-plane spin-orbit coupling,

these interactions are isotropic and favor spin alignment between the atoms. We may thus employ a theoretical treatment that assumes atoms of each horizontal plane are locked into a single large collective spin.

These large spins undergo both off-site and on-site spin-spin interactions (see Fig. S6A). The off-site interactions between neighboring vertical planes stem from s -wave collisions, which are enabled by the spin anisotropy introduced by the clock SOC. The same s -wave collisions cannot generate spatially anisotropic in-plane interactions due to the fermionic exchange statistics of the atoms, which mandates that same-spin atoms occupy antisymmetric spatial wavefunctions. Hence, s -wave collisions cannot occur if these atoms behave collectively (with symmetric spin wavefunctions). However, for $V_{\perp} \approx 0$ there can be non-negligible p -wave collisions, which act on-site between different radial modes and compete with the off-site interactions. The precise details of this description can be found in (19) and its supplemental information. An overview of its mathematical treatment for the system investigated in this work is also provided in the following section *1D large-spin model*.

The observed T_2 coherence time and atom lifetime for $V_{\perp} = 0$ are plotted as a function of V_z in Fig. S6B. Varying V_z provides two distinct regimes to probe the physics of contrast decay. At large V_z , off-site interactions are suppressed and the atoms predominantly interact via on-site p -wave interactions that contribute to contrast decay, as observed in previous studies (15). As V_z decreases, the p -wave interactions are reduced, leading to a slower decoherence rate. However, further decreasing V_z also leads to the lattice Wannier-Stark states along z becoming more delocalized and thus amplifying off-site s -wave interactions, which lead to an increasing decoherence rate. The crossover between these two mechanisms occurs around $V_z = 17.4E_R$, with a correspondingly longest coherence time of 19(5) s. While experiment and theory largely agree with each other, the discrepancy at long coherence times could arise from an unexpected reduction of the s -wave interaction strength from heating processes neglected in the theory. Note that the experiment in this work operates at a much higher density than previous studies (1, 19), thus the atom lifetimes (see Fig. S6B inset), limited by inelastic p -wave loss, are correspondingly much shorter (1).

Upon introduction of a weak transverse confinement ($V_{\perp} \neq 0$ but $V_{\perp} \ll V_z$), atoms are increasingly localized within each x - y plane, which leads to an enhancement of off-site s -wave interactions and suppression of on-site p -wave interactions. The latter substantially improves atom lifetimes, as shown in Fig. S6C. The T_2 coherence time in this weak transverse confinement regime exhibits different trends between the intermediate $V_z = 23.2E_R$ and deep $V_z = 46.4E_R$ lattices. For intermediate $V_z = 23.2E_R$ we find increasing transverse confinement $V_{\perp}$ lowers T_2 , as off-site s -wave interactions that generate undesirable spin-wave excitations are non-negligible and grow with increasing $V_{\perp}$. For

deeper $V_z = 46.4E_R$ the trend is reversed, and T_2 improves with increasing $V_\perp$, as in this latter regime the off-site s -wave interactions are sufficiently weak that any amplification of them by increasing $V_\perp$ still remains negligible. In this case T_2 is still limited by p -wave interactions, which are suppressed for larger $V_\perp$.

1D large-spin model

To theoretically model the contrast decay in Fig. S6 we describe the $V_\perp \ll V_z$ regime using the assumption that the spins in each pancake are locked into a large spin, based on Ref. (15, 19), which leads to the following 1D large-spin Hamiltonian:

$$\begin{aligned}\hat{H}_{\text{LS}} &= \hat{H}_{\text{on-site}} + \hat{H}_{\text{off-site}}, \\ \hat{H}_{\text{on-site}}/\hbar &= \sum_n \left[J_{0,n} \hat{\mathbf{S}}_n \cdot \hat{\mathbf{S}}_n + \chi_{0,n} \hat{S}_n^Z \hat{S}_n^Z + C_{0,n} \hat{N}_n \hat{S}_n^Z \right], \\ \hat{H}_{\text{off-site}}/\hbar &= \sum_n \left[J_{1,n} \hat{\mathbf{S}}_n \cdot \hat{\mathbf{S}}_{n+1} + \chi_{1,n} \hat{S}_n^Z \hat{S}_{n+1}^Z + D_{1,n} (\hat{S}_n^X \hat{S}_{n+1}^Y - \hat{S}_n^Y \hat{S}_{n+1}^X) \right].\end{aligned}\tag{S20}$$

The collective spin operators are defined as $\hat{\mathbf{S}}_n = \sum_{n_x n_y} \sum_{\alpha\beta=\{e,g\}} \hat{c}_{n_x n_y n, \alpha}^\dagger \boldsymbol{\sigma}_{\alpha\beta} \hat{c}_{n_x n_y n, \beta} / 2$ in the lab frame, where $\hat{c}_{n_x n_y n, \alpha}$ annihilate fermions in internal state α , Wannier-Stark level n , and radial mode (n_x, n_y) of the in-plane harmonic confinement (assumed to be a separable potential). Here $\hat{N}_n = \sum_{n_x n_y} \sum_{\alpha=\{e,g\}} \hat{c}_{n_x n_y n, \alpha}^\dagger \hat{c}_{n_x n_y n, \alpha}$ is the atom number operator. As in the 3D case, the spin Hamiltonian is written in a “spiral” frame that absorbs the spin-orbit coupling phase, with $\hat{S}_n^\pm = e^{\pm i n \varphi} \hat{S}_n^\pm$ and $\hat{S}_n^z = \hat{S}_n^z$. The interaction parameters are

$$\begin{aligned}J_{0,n} &= \eta_0 (V_{eg}^{n,n} - U_{eg}^{n,n}) / 2, \quad \chi_{0,n} = \eta_0 (V_{ee}^{n,n} + V_{gg}^{n,n} - 2V_{eg}^{n,n}) / 2, \\ C_{0,n} &= \eta_0 (V_{ee}^{n,n} - V_{gg}^{n,n}) / 2, \quad J_{1,n} = -\eta_1 U_{eg}^{n,n+1} \cos \varphi, \\ \chi_{1,n} &= -\eta_1 U_{eg}^{n,n+1} (1 - \cos \varphi), \quad D_{1,n} = -\eta_1 U_{eg}^{n,n+1} \sin \varphi.\end{aligned}\tag{S21}$$

where η_0 and η_1 are dimensionless overlap integrals,

$$\eta_{|n-m|} = \frac{\sqrt{2\pi}}{k_L} \left(\frac{V_z}{E_R} \right)^{-1/4} \int dz [W_n(z)]^2 [W_m(z)]^2,\tag{S22}$$

with $W_n(z)$ the wavefunction of a Wannier-Stark state centered at site n . The s -wave ($U_{\alpha\beta}$) and p -wave ($V_{\alpha\beta}$) interaction strengths ($\alpha, \beta = \{g, e\}$) are calculated by averaging

a Fermi-Dirac distribution over the radial modes,

$$\begin{aligned}
U_{\alpha\beta}^{n,m} &= \frac{8\pi\hbar a_{\alpha\beta}}{m} \frac{k_L}{\sqrt{2\pi}} \left(\frac{V_z}{E_R}\right)^{1/4} \sum_{n_x m_x n_y m_y} s_{n_x m_x} s_{n_y m_y} \frac{N_{n_x n_y n}}{N_{n,\text{init}}} \frac{N_{m_x m_y m}}{N_{m,\text{init}}}, \\
V_{\alpha\beta}^{n,m} &= \frac{6\pi\hbar b_{\alpha\beta}^3}{m} \frac{k_L}{\sqrt{2\pi}} \left(\frac{V_z}{E_R}\right)^{1/4} \sum_{n_x m_x n_y m_y} (p_{n_x m_x} s_{n_y m_y} + s_{n_x m_x} p_{n_y m_y}) \frac{N_{n_x n_y n}}{N_{n,\text{init}}} \frac{N_{m_x m_y m}}{N_{m,\text{init}}},
\end{aligned} \tag{S23}$$

where $a_{\alpha\beta}$ is the elastic s -wave scattering length, and $b_{\alpha\beta}^3$ is the elastic p -wave scattering volume. As the atoms are nuclear-spin polarized, for $e - g$ collisions we require the specific scattering lengths $a_{eg} = a_{eg}^-$, $b_{eg}^3 = (b_{eg}^+)^3$ to maintain the correct symmetry (16, 19). Defining the wave function for a radial mode (n_x, n_y) is $\phi_{n_x} \phi_{n_y}$, we have $s_{nm} = \int dx [\phi_n(x)]^2 [\phi_m(x)]^2$, $p_{nm} = \int dx [(\partial_x \phi_n(x)) \phi_m(x) - \phi_n(x) (\partial_x \phi_m(x))]^2$. Here $N_{n_x n_y n}$ are the initial populations in radial mode (n_x, n_y) and lattice site n under a Fermi-Dirac distribution, $N_{n_x n_y n} = \left[\exp[(\epsilon_{n_x n_y} - \mu_n)/k_B T_R] + 1 \right]^{-1}$. The chemical potential for each lattice site μ_n is set by the initial atom number $N_{n,\text{init}} = \sum_{n_x n_y} N_{n_x n_y n}$, while T_R is the radial temperature.

For $V_\perp = 0$, the radial modes $\phi_{n_x} \phi_{n_y}$ are taken to be 2D harmonic oscillator modes with trapping frequency set by the beam waist of the vertical lattice beam. We set the radial temperature to $T_R(\text{nK}) = 60 \times \sqrt{V_z/E_R}$, obtained by calibrated comparison of the density distribution between theory and experiment. In the case of $V_\perp > 0$, the radial modes are instead set to eigenfunctions of both the harmonic confinement from the vertical beam and the 2D in-plane lattice. For the latter, the compression from ramping up the lattice in the loading sequence leads to a lower temperature compared to the case of $V_\perp = 0$. Therefore, as radial temperature we use $T_R(V_\perp > 0) = 0.42 \times T_R(V_\perp = 0)$ obtained from a fit of the simulated T_2 values to the experimental data to generate the atom distribution in radial modes.

Apart from unitary evolution under $\hat{H}_{\text{LS}}$, inelastic on-site p -wave e - e collision can lead to two-body loss of the atom number as observed in previous studies (15, 16). We describe such atom loss based on the following Lindblad master equation,

$$\begin{aligned}
\hbar \frac{d}{dT} \hat{\rho} &= -i[\hat{H}_{\text{LS}}, \hat{\rho}] + \sum_n \Gamma_{0,n} \mathcal{L}_n(\hat{\rho}), \\
\mathcal{L}_n(\hat{\rho}) &= \sum_{n_x n_y m_x m_y} \left[\hat{L}_{n_x n_y m_x m_y} \hat{\rho} \hat{L}_{n_x n_y m_x m_y}^\dagger - \frac{1}{2} \{ \hat{L}_{n_x n_y m_x m_y}^\dagger \hat{L}_{n_x n_y m_x m_y}, \hat{\rho} \} \right],
\end{aligned} \tag{S24}$$

where $\hat{L}_{n_x n_y m_x m_y} = \hat{c}_{n_x n_y n, e} \hat{c}_{m_x m_y n, e}$, and the loss rates are averaged over the radial

modes,

$$\Gamma_{0,n} = \eta_0 \tilde{V}_{ee}^{n,n}/4, \quad (\text{S25})$$

where $\tilde{V}_{ee}^{n,n}$ is $V_{ee}^{n,n}$ from Eq. (S23) with the elastic p -wave scattering volume b_{ee}^3 replaced by the inelastic p -wave scattering volume β_{ee}^3 .

This master equation is solved numerically via the truncated Wigner approximation (TWA) (58) by solving the mean-field equations of Eq. (S24) with random sampling of initial conditions. We set $S_n^X(0) = N_{n,\text{init}}/2$, $N_n(0) = N_{n,\text{init}}$, and sample $S_n^Y(0)$ and $S_n^Z(0)$ using a Gaussian distribution of mean zero and variance $\sigma^2 = N_{n,\text{init}}/2$.

In Fig. S6 the red lines for the T_2 value and atom number lifetime are extracted from the crossing of the $1/e$ value in the transverse spin amplitude and atom number obtained from truncated Wigner simulations based on the model discussed above. Errorbands in the same figure include the uncertainty of s -wave and p -wave scattering parameters, $0.5E_R$ uncertainty of lattice depth, as well as 20% uncertainty in radial temperature.

Note that if we assume the atom loss for ground and excited states is symmetric due to the XY8 pulses, we have a simplified equation for the atomic population N_n specifically,

$$\frac{d}{dT}N_n = -\frac{\Gamma_{0,n}}{2}N_n^2, \quad (\text{S26})$$

which gives an exact solution

$$N_n(T) = \frac{N_{n,\text{init}}}{1 + \Gamma_{0,n}N_{n,\text{init}}T/2}. \quad (\text{S27})$$

We extract the total atom number measured in the experiment (black points in Fig. S6) by integrating through all lattice layers, then using the fitting function $A/(1 + BT)$ with fitting parameters A, B to extract the atom loss rate. We confirmed that this approach is consistent with the extraction method applied in the simulations.

$V_{\perp} [E_R]$	N	ω_{lat}/h [Hz]	$j_0 - j_s [a]$	U/h [kHz]	T_2 [s]
19.6	22,100	12.4	35	1.17	5.7
22.5	13,000	14.3	31	1.27	6.9
28.1	14,800	17.8	25	1.44	9.1
33.7	17,700	21.4	20	1.59	11.4
44.9	22,500	28.5	15	1.86	10.4

Table S1: **Parameters for 3D lattice simulations.** For transverse lattice depth $V_{\perp}$ we provide the atom number N , the lattice harmonic confinement frequency ω_{lat} along the z direction, the position of the cloud $j_0 - j_s$ relative to the minimum of the lattice confinement (in units of lattice spacing), the on-site Hubbard repulsion U at the center of the lattice, and the lifetime of atomic coherence T_2 (beyond decay caused by superexchange).

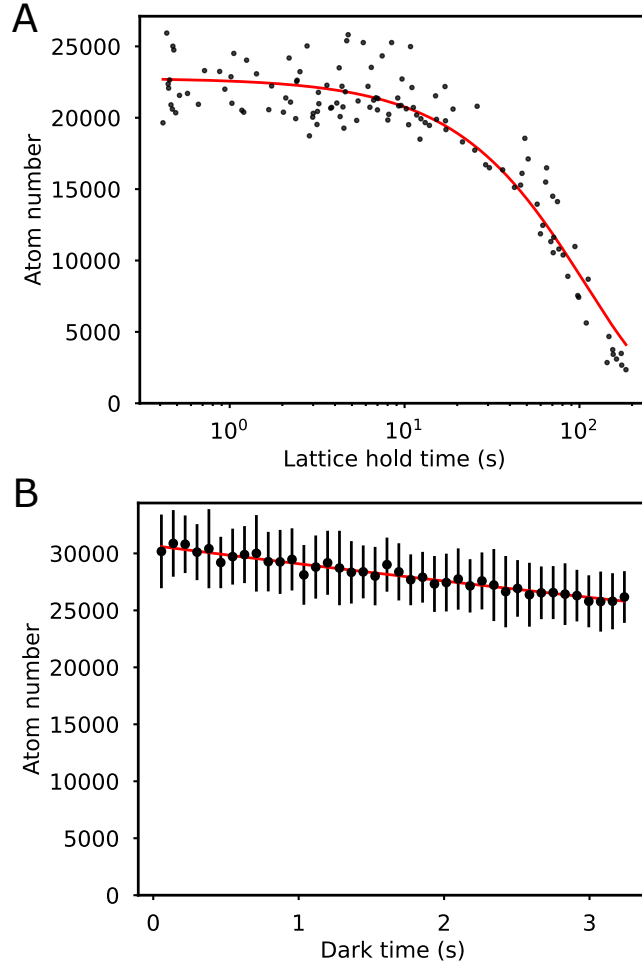


Figure S1: **Atom lifetime measurements.** **A** To determine the single-particle lifetime in the 3D lattice, we load ground state atoms in 1S_0 into a deep lattice. Applying an exponential fit (red line), a $1/e$ time of 108(5) seconds is fit to the data. In this measurement, $V_{\perp,1} = 66.5E_R$, $V_{\perp,2} = 69.9E_R$, $V_z = 43.7E_R$. The atom loss is likely limited by a combination of parametric heating from trap and the vacuum lifetime. This loss timescale is much longer than all dynamics studied in this work. **B** In the superexchange regime $V_{\perp} = 44.9E_R$ and $V_z = 17.4E_R$, we fit a $1/e$ time constant $\tau = 19(1)$ seconds. We emphasize that in this regime atoms are in a superposition state of 1S_0 and 3P_0 and thus susceptible to e-g collisional loss. Still, the loss rate in the superexchange regime is $\approx 10\times$ slower than the superexchange rate J_{SE} . Statistical errorbars in both of these measurements are susceptible to long term drifts of the total atom number.

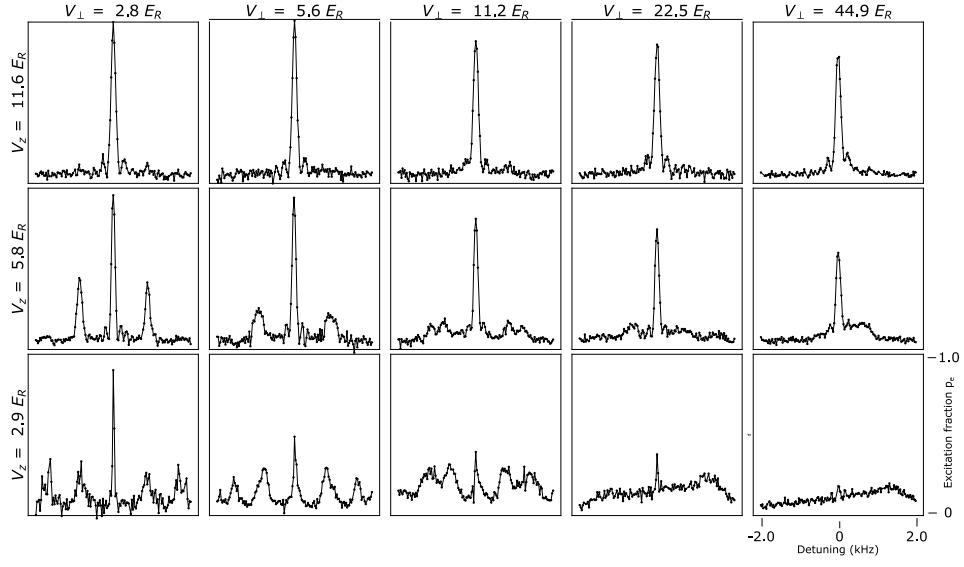


Figure S2: **Shallow lattice Rabi spectroscopy.** Rabi spectroscopy in shallow lattice confinement is studied. The detuning span in each measurement is -2kHz to 2kHz. The Rabi pulse duration was optimized for each measurement to maximize the excited state fraction at zero detuning.

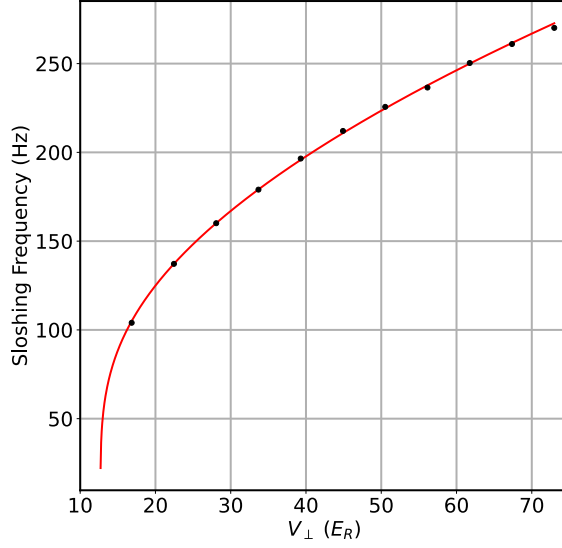


Figure S3: **Lattice Gaussian confinement calibration.** The trap frequency is measured for atoms solely confined in both horizontal lattice beams, each at a power equivalent to the trap depth $V_{\perp}$. The oscillations are initiated by rapidly switching off the superimposed horizontal dipole trap. We note, that for the evaluation the gravitational sag needs to be taken into account, and that employing a harmonic approximation for estimating the sag induces discernible errors in the beam parameters. Numerical evaluation of the Gaussian beam curvature at the position of the atomic sag allows the construction of a fit function from which the beam waist radius is extracted to be $w = 62.2(14) \mu\text{m}$.

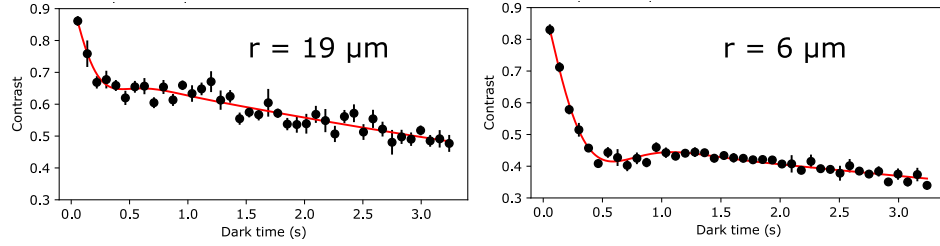


Figure S4: **Radial dependence on contrast.** Contrast decay in two regions of interest using annuli with thickness 2 px ($0.8 \mu\text{m}$) and radii 14 px ($6 \mu\text{m}$), 48 px ($19 \mu\text{m}$) are plotted. $V_{\perp} = 44.9 E_R$ and $V_Z = 17.4 E_R$ was used for these measurements. Error bars are 1σ (standard deviation) uncertainty of the Ramsey contrast obtained from jackknifing. The sparsely filled region ($19 \mu\text{m}$) has higher contrast due to the lower filling fraction compared to near the center of the cloud ($6 \mu\text{m}$).

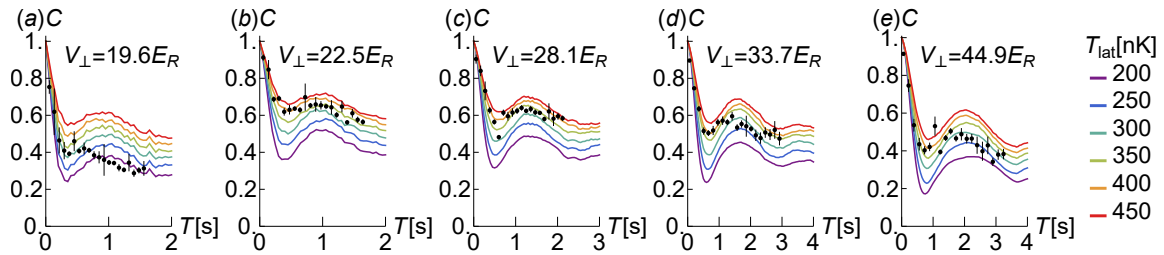


Figure S5: **Contrast temperature dependence.** Contrast decay theoretically predicted using full many-body simulations and averaging over experimental conditions. The parameters used are taken from Table S1.

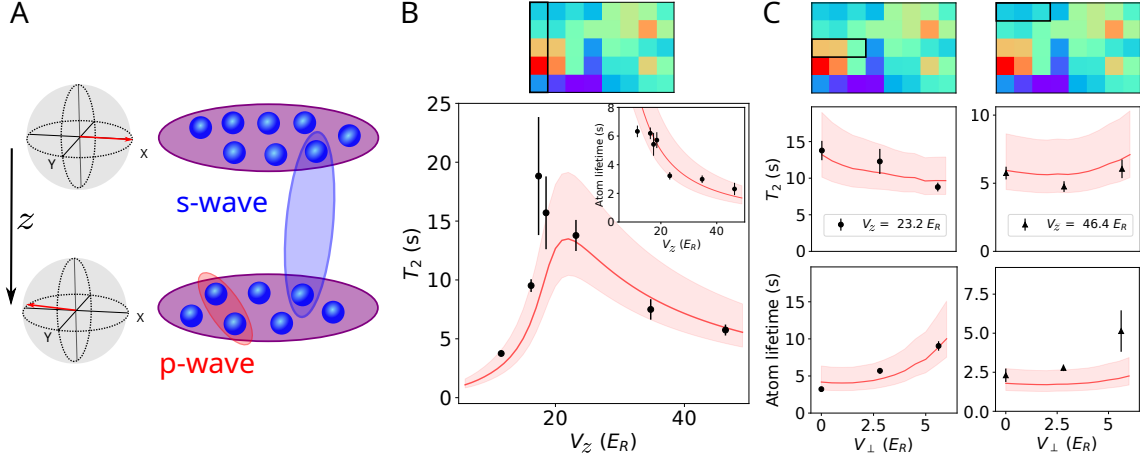


Figure S6: Weak transverse confinement regime. **A**, In the weak transverse confinement regime, both off-site s -wave interactions, induced by the SOC phase between lattice sites, and on-site p -wave interactions between atoms contribute to dephasing (19). Their strength is controlled by the vertical confinement V_z and transverse confinement $V_\perp$, strongly influencing the observed coherence time T_2 . **B**, T_2 is measured without transverse confinement ($V_\perp = 0$). In the inset, the atom lifetime τ , limited by inelastic p -wave loss, is plotted as a function of V_z . Theory modeling Ramsey contrast decay based on the 1D spin Hamiltonian using experimental measured parameters is overlaid in red. The error bands are based on the uncertainties of the experimental parameters. Error bars are 1σ (standard deviation) uncertainty of the fitted T_2 and τ values. **C**, A weak transverse confinement $V_\perp$ is applied. This leads to increased τ , as well as a reduction of T_2 at intermediate $V_z = 23.2E_R$. In contrast, at deep $V_z = 46.4E_R$ the coherence time is enhanced with $V_\perp$, although the values are smaller than the ones at $V_z = 23.2E_R$. The reproductions of the quality factor map from Fig. 2C at the top of this figure indicate the lattice potential range investigated in the corresponding subfigures.

References and Notes

1. T. Bothwell, C. J. Kennedy, A. Aepli, D. Kedar, J. M. Robinson, E. Oelker, A. Staron, J. Ye, Resolving the gravitational redshift across a millimetre-scale atomic sample. *Nature* **602**, 420–424 (2022). [doi:10.1038/s41586-021-04349-7](https://doi.org/10.1038/s41586-021-04349-7) [Medline](#)
2. A. Aepli, K. Kim, W. Warfield, M. S. Safronova, J. Ye, Clock with 8×10^{-19} Systematic Uncertainty. *Phys. Rev. Lett.* **133**, 023401 (2024). [doi:10.1103/PhysRevLett.133.023401](https://doi.org/10.1103/PhysRevLett.133.023401) [Medline](#)
3. S. Kolkowitz, I. Pikovski, N. Langellier, M. D. Lukin, R. L. Walsworth, J. Ye, Gravitational wave detection with optical lattice atomic clocks. *Phys. Rev. D* **94**, 124043 (2016). [doi:10.1103/PhysRevD.94.124043](https://doi.org/10.1103/PhysRevD.94.124043)
4. R. B. Hutson, W. R. Milner, L. Yan, J. Ye, C. Sanner, Observation of millihertz-level cooperative Lamb shifts in an optical atomic clock. *Science* **383**, 384–387 (2024). [doi:10.1126/science.adh4477](https://doi.org/10.1126/science.adh4477) [Medline](#)
5. Boulder Atomic Clock Optical Network (BACON) Collaboration, Frequency ratio measurements at 18-digit accuracy using an optical clock network. *Nature* **591**, 564–569 (2021). [doi:10.1038/s41586-021-03253-4](https://doi.org/10.1038/s41586-021-03253-4) [Medline](#)
6. N. Nemitz, T. Ohkubo, M. Takamoto, I. Ushijima, M. Das, N. Ohmae, H. Katori, Frequency ratio of Yb and Sr clocks with 5×10^{-17} uncertainty at 150 seconds averaging time. *Nat. Photonics* **10**, 258–261 (2016). [doi:10.1038/nphoton.2016.20](https://doi.org/10.1038/nphoton.2016.20)
7. W. F. McGrew, X. Zhang, R. J. Fasano, S. A. Schäffer, K. Beloy, D. Nicolodi, R. C. Brown, N. Hinkley, G. Milani, M. Schioppo, T. H. Yoon, A. D. Ludlow, Atomic clock performance enabling geodesy below the centimetre level. *Nature* **564**, 87–90 (2018). [doi:10.1038/s41586-018-0738-2](https://doi.org/10.1038/s41586-018-0738-2) [Medline](#)
8. W. M. Itano, J. C. Bergquist, J. J. Bollinger, J. M. Gilligan, D. J. Heinzen, F. L. Moore, M. G. Raizen, D. J. Wineland, Quantum projection noise: Population fluctuations in two-level systems. *Phys. Rev. A* **47**, 3554–3570 (1993). [doi:10.1103/PhysRevA.47.3554](https://doi.org/10.1103/PhysRevA.47.3554) [Medline](#)
9. E. Pedrozo-Peñafiel, S. Colombo, C. Shu, A. F. Adiyatullin, Z. Li, E. Mendez, B. Braverman, A. Kawasaki, D. Akamatsu, Y. Xiao, V. Vuletić, Entanglement on an optical atomic-clock transition. *Nature* **588**, 414–418 (2020). [doi:10.1038/s41586-020-3006-1](https://doi.org/10.1038/s41586-020-3006-1) [Medline](#)
10. J. M. Robinson, M. Miklos, Y. M. Tso, C. J. Kennedy, T. Bothwell, D. Kedar, J. K. Thompson, J. Ye, Direct comparison of two spin squeezed optical clock ensembles at the 10^{-17} level. *Nat. Phys.* **20**, 208–213 (2024). [doi:10.1038/s41567-023-02310-1](https://doi.org/10.1038/s41567-023-02310-1)
11. W. J. Eckner, N. Darkwah Oppong, A. Cao, A. W. Young, W. R. Milner, J. M. Robinson, J. Ye, A. M. Kaufman, Realizing spin squeezing with Rydberg interactions in an optical clock. *Nature* **621**, 734–739 (2023). [doi:10.1038/s41586-023-06360-6](https://doi.org/10.1038/s41586-023-06360-6) [Medline](#)
12. J. Franke, S. R. Muleady, R. Kaubuegger, F. Kranzl, R. Blatt, A. M. Rey, M. K. Joshi, C. F. Roos, Quantum-enhanced sensing on optical transitions through finite-range interactions. *Nature* **621**, 740–745 (2023). [doi:10.1038/s41586-023-06472-z](https://doi.org/10.1038/s41586-023-06472-z) [Medline](#)
13. E. Oelker, R. B. Hutson, C. J. Kennedy, L. Sonderhouse, T. Bothwell, A. Goban, D. Kedar, C. Sanner, J. M. Robinson, G. E. Marti, D. G. Matei, T. Legero, M. Giunta, R.

- Holzwarth, F. Riehle, U. Sterr, J. Ye, Demonstration of 4.8×10^{-17} stability at 1 s for two independent optical clocks. *Nat. Photonics* **13**, 714–719 (2019). [doi:10.1038/s41566-019-0493-4](https://doi.org/10.1038/s41566-019-0493-4)
14. A. D. Ludlow, M. M. Boyd, J. Ye, E. Peik, P. O. Schmidt, Optical atomic clocks. *Rev. Mod. Phys.* **87**, 637–701 (2015). [doi:10.1103/RevModPhys.87.637](https://doi.org/10.1103/RevModPhys.87.637)
 15. M. J. Martin, M. Bishof, M. D. Swallows, X. Zhang, C. Benko, J. von-Stecher, A. V. Gorshkov, A. M. Rey, J. Ye, A quantum many-body spin system in an optical lattice clock. *Science* **341**, 632–636 (2013). [doi:10.1126/science.1236929](https://doi.org/10.1126/science.1236929) [Medline](#)
 16. X. Zhang, M. Bishof, S. L. Bromley, C. V. Kraus, M. S. Safronova, P. Zoller, A. M. Rey, J. Ye, Quantum simulation. Spectroscopic observation of SU(N)-symmetric interactions in Sr orbital magnetism. *Science* **345**, 1467–1473 (2014). [doi:10.1126/science.1254978](https://doi.org/10.1126/science.1254978) [Medline](#)
 17. S. Kolkowitz, S. L. Bromley, T. Bothwell, M. L. Wall, G. E. Marti, A. P. Koller, X. Zhang, A. M. Rey, J. Ye, Spin-orbit-coupled fermions in an optical lattice clock. *Nature* **542**, 66–70 (2017). [doi:10.1038/nature20811](https://doi.org/10.1038/nature20811) [Medline](#)
 18. S. L. Bromley, S. Kolkowitz, T. Bothwell, D. Kedar, A. Safavi-Naini, M. L. Wall, C. Salomon, A. M. Rey, J. Ye, Dynamics of interacting fermions under spin-orbit coupling in an optical lattice clock. *Nat. Phys.* **14**, 399–404 (2018). [doi:10.1038/s41567-017-0029-0](https://doi.org/10.1038/s41567-017-0029-0)
 19. A. Aepli, A. Chu, T. Bothwell, C. J. Kennedy, D. Kedar, P. He, A. M. Rey, J. Ye, Hamiltonian engineering of spin-orbit-coupled fermions in a Wannier-Stark optical lattice clock. *Sci. Adv.* **8**, eadc9242 (2022). [doi:10.1126/sciadv.adc9242](https://doi.org/10.1126/sciadv.adc9242) [Medline](#)
 20. S. L. Campbell, R. B. Hutson, G. E. Marti, A. Goban, N. Darkwah Oppong, R. L. McNally, L. Sonderhouse, J. M. Robinson, W. Zhang, B. J. Bloom, J. Ye, A Fermi-degenerate three-dimensional optical lattice clock. *Science* **358**, 90–94 (2017). [doi:10.1126/science.aam5538](https://doi.org/10.1126/science.aam5538) [Medline](#)
 21. A. Goban, R. B. Hutson, G. E. Marti, S. L. Campbell, M. A. Perlin, P. S. Julianne, J. P. D’Incao, A. M. Rey, J. Ye, Emergence of multi-body interactions in a fermionic lattice clock. *Nature* **563**, 369–373 (2018). [doi:10.1038/s41586-018-0661-6](https://doi.org/10.1038/s41586-018-0661-6) [Medline](#)
 22. R. B. Hutson, A. Goban, G. E. Marti, L. Sonderhouse, C. Sanner, J. Ye, Engineering quantum states of matter for atomic clocks in shallow optical lattices. *Phys. Rev. Lett.* **123**, 123401 (2019). [doi:10.1103/PhysRevLett.123.123401](https://doi.org/10.1103/PhysRevLett.123.123401) [Medline](#)
 23. L. Sonderhouse, C. Sanner, R. B. Hutson, A. Goban, T. Bilitewski, L. Yan, W. R. Milner, A. M. Rey, J. Ye, Thermodynamics of a deeply degenerate SU(N)-symmetric Fermi gas. *Nat. Phys.* **16**, 1216–1221 (2020). [doi:10.1038/s41567-020-0986-6](https://doi.org/10.1038/s41567-020-0986-6)
 24. L.-M. Duan, E. Demler, M. D. Lukin, Controlling spin exchange interactions of ultracold atoms in optical lattices. *Phys. Rev. Lett.* **91**, 090402 (2003). [doi:10.1103/PhysRevLett.91.090402](https://doi.org/10.1103/PhysRevLett.91.090402) [Medline](#)
 25. M. Lewenstein, A. Sanpera, V. Ahufinger, B. Damski, A. Sen(De), U. Sen, Ultracold atomic gases in optical lattices: Mimicking condensed matter physics and beyond. *Adv. Phys.* **56**, 243–379 (2007). [doi:10.1080/00018730701223200](https://doi.org/10.1080/00018730701223200)

26. S. Trotzky, P. Cheinet, S. Fölling, M. Feld, U. Schnorrberger, A. M. Rey, A. Polkovnikov, E. A. Demler, M. D. Lukin, I. Bloch, Time-resolved observation and control of superexchange interactions with ultracold atoms in optical lattices. *Science* **319**, 295–299 (2008). [doi:10.1126/science.1150841](https://doi.org/10.1126/science.1150841) [Medline](#)
27. E. Manousakis, The spin- $\frac{1}{2}$ Heisenberg antiferromagnet on a square lattice and its application to the cuprous oxides. *Rev. Mod. Phys.* **63**, 1–62 (1991). [doi:10.1103/RevModPhys.63.1](https://doi.org/10.1103/RevModPhys.63.1)
28. A. Auerbach, *Interacting Electrons and Quantum Magnetism* (Springer, 2012); <https://link.springer.com/book/10.1007/978-1-4612-0869-3>.
29. P. A. Lee, N. Nagaosa, X.-G. Wen, Doping a Mott insulator: Physics of high-temperature superconductivity. *Rev. Mod. Phys.* **78**, 17–85 (2006). [doi:10.1103/RevModPhys.78.17](https://doi.org/10.1103/RevModPhys.78.17)
30. D. Greif, T. Uehlinger, G. Jotzu, L. Tarruell, T. Esslinger, Short-range quantum magnetism of ultracold fermions in an optical lattice. *Science* **340**, 1307–1310 (2013). [doi:10.1126/science.1236362](https://doi.org/10.1126/science.1236362) [Medline](#)
31. R. A. Hart, P. M. Duarte, T.-L. Yang, X. Liu, T. Paiva, E. Khatami, R. T. Scalettar, N. Trivedi, D. A. Huse, R. G. Hulet, Observation of antiferromagnetic correlations in the Hubbard model with ultracold atoms. *Nature* **519**, 211–214 (2015). [doi:10.1038/nature14223](https://doi.org/10.1038/nature14223) [Medline](#)
32. M. Boll, T. A. Hilker, G. Salomon, A. Omran, J. Nespolo, L. Pollet, I. Bloch, C. Gross, Spin- and density-resolved microscopy of antiferromagnetic correlations in Fermi-Hubbard chains. *Science* **353**, 1257–1260 (2016). [doi:10.1126/science.aag1635](https://doi.org/10.1126/science.aag1635) [Medline](#)
33. L. W. Cheuk, M. A. Nichols, K. R. Lawrence, M. Okan, H. Zhang, E. Khatami, N. Trivedi, T. Paiva, M. Rigol, M. W. Zwierlein, Observation of spatial charge and spin correlations in the 2D Fermi-Hubbard model. *Science* **353**, 1260–1264 (2016). [doi:10.1126/science.aag3349](https://doi.org/10.1126/science.aag3349) [Medline](#)
34. A. Mazurenko, C. S. Chiu, G. Ji, M. F. Parsons, M. Kanász-Nagy, R. Schmidt, F. Grusdt, E. Demler, D. Greif, M. Greiner, A cold-atom Fermi-Hubbard antiferromagnet. *Nature* **545**, 462–466 (2017). [doi:10.1038/nature22362](https://doi.org/10.1038/nature22362) [Medline](#)
35. M. Takahashi, *Thermodynamics of One-Dimensional Solvable Models* (Cambridge Univ. Press, 2005); <https://www.cambridge.org/core/books/thermodynamics-of-one-dimensional-solvable-models/B74C2EB63A89A1E5D7347EE48E4AFD6C>.
36. S. Taie, E. Ibarra-García-Padilla, N. Nishizawa, Y. Takasu, Y. Kuno, H.-T. Wei, R. T. Scalettar, K. R. A. Hazzard, Y. Takahashi, Observation of antiferromagnetic correlations in an ultracold SU (N) Hubbard model. *Nat. Phys.* **18**, 1356–1361 (2022). [doi:10.1038/s41567-022-01725-6](https://doi.org/10.1038/s41567-022-01725-6)
37. M. Gall, N. Wurz, J. Samland, C. F. Chan, M. Köhl, Competing magnetic orders in a bilayer Hubbard model with ultracold atoms. *Nature* **589**, 40–43 (2021). [doi:10.1038/s41586-020-03058-x](https://doi.org/10.1038/s41586-020-03058-x) [Medline](#)
38. R. C. Brown, R. Wyllie, S. B. Koller, E. A. Goldschmidt, M. Foss-Feig, J. V. Porto, Quantum simulation. Two-dimensional superexchange-mediated magnetization dynamics in an optical lattice. *Science* **348**, 540–544 (2015). [doi:10.1126/science.aaa1385](https://doi.org/10.1126/science.aaa1385) [Medline](#)

39. P. N. Jepsen, J. Amato-Grill, I. Dimitrova, W. W. Ho, E. Demler, W. Ketterle, Spin transport in a tunable Heisenberg model realized with ultracold atoms. *Nature* **588**, 403–407 (2020). [doi:10.1038/s41586-020-3033-y](https://doi.org/10.1038/s41586-020-3033-y) [Medline](#)
40. H. Sun, B. Yang, H.-Y. Wang, Z.-Y. Zhou, G.-X. Su, H.-N. Dai, Z.-S. Yuan, J.-W. Pan, Realization of a bosonic antiferromagnet. *Nat. Phys.* **17**, 990–994 (2021). [doi:10.1038/s41567-021-01277-1](https://doi.org/10.1038/s41567-021-01277-1)
41. M. Mamaev, I. Kimchi, R. M. Nandkishore, A. M. Rey, Tunable-spin-model generation with spin-orbit-coupled fermions in optical lattices. *Phys. Rev. Res.* **3**, 013178 (2021). [doi:10.1103/PhysRevResearch.3.013178](https://doi.org/10.1103/PhysRevResearch.3.013178)
42. P. N. Jepsen, Y. K. E. Lee, H. Lin, I. Dimitrova, Y. Margalit, W. W. Ho, W. Ketterle, Long-lived phantom helix states in Heisenberg quantum magnets. *Nat. Phys.* **18**, 899–904 (2022). [doi:10.1038/s41567-022-01651-7](https://doi.org/10.1038/s41567-022-01651-7)
43. P. He, M. A. Perlin, S. R. Muleady, R. J. Lewis-Swan, R. B. Hutson, J. Ye, A. M. Rey, Engineering spin squeezing in a 3D optical lattice with interacting spin-orbit-coupled fermions. *Phys. Rev. Res.* **1**, 033075 (2019). [doi:10.1103/PhysRevResearch.1.033075](https://doi.org/10.1103/PhysRevResearch.1.033075)
44. T. Hernández Yanes, M. Płodzień, M. Mackoīt Sinkevičienė, G. Žlabys, G. Juzeliūnas, E. Witkowska, One-and two-axis squeezing via laser coupling in an atomic Fermi-Hubbard model. *Phys. Rev. Lett.* **129**, 090403 (2022). [doi:10.1103/PhysRevLett.129.090403](https://doi.org/10.1103/PhysRevLett.129.090403) [Medline](#)
45. M. Mamaev, D. Barberena, A. M. Rey, Spin squeezing in mixed-dimensional anisotropic lattice models. *Phys. Rev. A* **109**, 023326 (2024). [doi:10.1103/PhysRevA.109.023326](https://doi.org/10.1103/PhysRevA.109.023326)
46. W. R. Milner, L. Yan, R. B. Hutson, C. Sanner, J. Ye, High-fidelity imaging of a band insulator in a three-dimensional optical lattice clock. *Phys. Rev. A* **107**, 063313 (2023). [doi:10.1103/PhysRevA.107.063313](https://doi.org/10.1103/PhysRevA.107.063313)
47. D. G. Matei, T. Legero, S. Häfner, C. Grebing, R. Weyrich, W. Zhang, L. Sonderhouse, J. M. Robinson, J. Ye, F. Riehle, U. Sterr, 1.5 μm lasers with sub-10 mHz linewidth. *Phys. Rev. Lett.* **118**, 263202 (2017). [doi:10.1103/PhysRevLett.118.263202](https://doi.org/10.1103/PhysRevLett.118.263202) [Medline](#)
48. G. E. Marti, R. B. Hutson, A. Goban, S. L. Campbell, N. Poli, J. Ye, Imaging optical frequencies with 100 μHz precision and 1.1 μm resolution. *Phys. Rev. Lett.* **120**, 103201 (2018). [doi:10.1103/PhysRevLett.120.103201](https://doi.org/10.1103/PhysRevLett.120.103201) [Medline](#)
49. T. Gullion, D. B. Baker, M. S. Conradi, New, compensated Carr-Purcell sequences. *J. Magn. Reson.* **89**, 479–484 (1990). [doi:10.1016/0022-2364\(90\)90331-3](https://doi.org/10.1016/0022-2364(90)90331-3)
50. J.-R. Li, K. Matsuda, C. Miller, A. N. Carroll, W. G. Tobias, J. S. Higgins, J. Ye, Tunable itinerant spin dynamics with polar molecules. *Nature* **614**, 70–74 (2023). [doi:10.1038/s41586-022-05479-2](https://doi.org/10.1038/s41586-022-05479-2) [Medline](#)
51. See supplementary materials.
52. T. Esslinger, Fermi-Hubbard physics with atoms in an optical lattice. *Annu. Rev. Condens. Matter Phys.* **1**, 129–152 (2010). [doi:10.1146/annurev-conmatphys-070909-104059](https://doi.org/10.1146/annurev-conmatphys-070909-104059)
53. P. Lemonde, P. Wolf, Optical lattice clock with atoms confined in a shallow trap. *Phys. Rev. A* **72**, 033409 (2005). [doi:10.1103/PhysRevA.72.033409](https://doi.org/10.1103/PhysRevA.72.033409)

- 54. M. Foss-Feig, K. R. A. Hazzard, J. J. Bollinger, A. M. Rey, Nonequilibrium dynamics of arbitrary-range Ising models with decoherence: An exact analytic solution. *Phys. Rev. A* **87**, 042101 (2013). [doi:10.1103/PhysRevA.87.042101](https://doi.org/10.1103/PhysRevA.87.042101)
- 55. A. L. Gaunt, T. F. Schmidutz, I. Gotlibovych, R. P. Smith, Z. Hadzibabic, Bose-Einstein condensation of atoms in a uniform potential. *Phys. Rev. Lett.* **110**, 200406 (2013). [doi:10.1103/PhysRevLett.110.200406](https://doi.org/10.1103/PhysRevLett.110.200406) [Medline](#)
- 56. W. R. Milner, S. Lannig, M. Mamaev, L. Yan, A. Chu, B. Lewis, M. N. Frankel, R. B. Hutson, A. M. Rey, J. Ye, Coherent evolution of superexchange interaction in seconds long optical clock spectroscopy, Data and code, Dryad (2024); <https://doi.org/10.5061/dryad.qv9s4mwq5>.
- 57. D.-S. Lühmann, O. Jürgensen, K. Sengstock, Multi-orbital and density-induced tunneling of bosons in optical lattices. *New J. Phys.* **14**, 033021 (2012). [doi:10.1088/1367-2630/14/3/033021](https://doi.org/10.1088/1367-2630/14/3/033021)
- 58. A. Polkovnikov, Phase space representation of quantum dynamics. *Ann. Phys.* **325**, 1790–1852 (2010). [doi:10.1016/j.aop.2010.02.006](https://doi.org/10.1016/j.aop.2010.02.006)